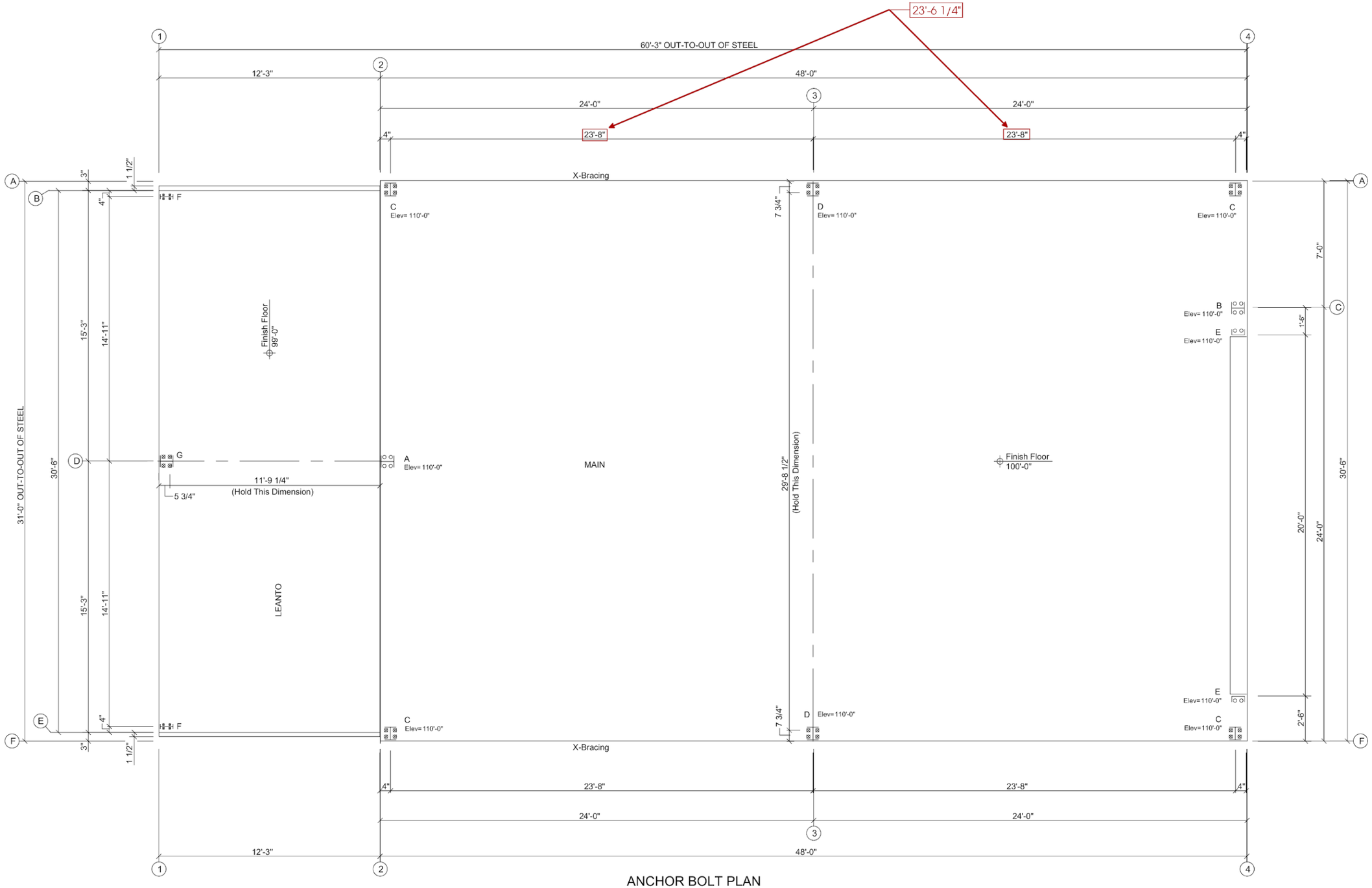


ANCHOR BOLT PLAN

NOTE: All Base Plates @ 100'-0" (U.N.)

Registered Professional Engineer (Structural)
CERTIFICATE NO.
77126
COLEMAN D. LARSEN
01/30/24
Date Signed
ARIZONA U.S.A.

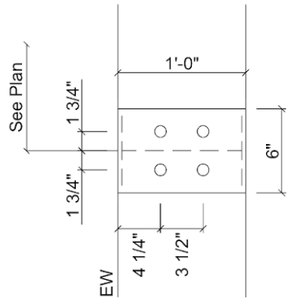
DRAWING REVISIONS	DRAWING USAGE
	For Construction 1/26/2024

IMAGE BUILDING SYSTEMS

	ANCHOR BOLT PLAN		IM-0017-23
CUSTOMER	ADOT DE-ICING	PROJECT	ADOT DE-ICING
	SUPERIOR, AZ 85173		PRJ Address 1 SUPERIOR, AZ 85173

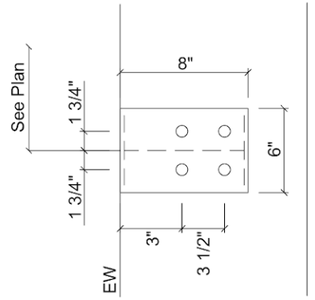
SHEET

Dia= 5/8"



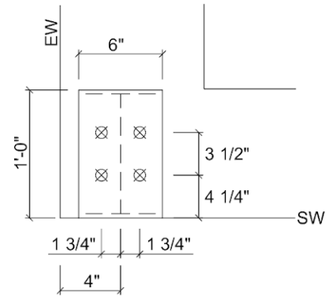
DETAIL A Base EL. 110'-0"

Dia= 5/8"



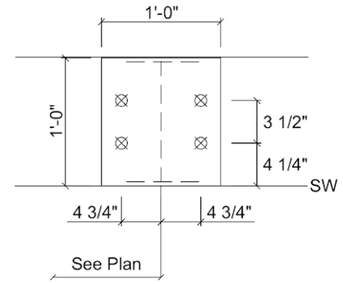
DETAIL B Base EL. 110'-0"

Dia= 3/4"



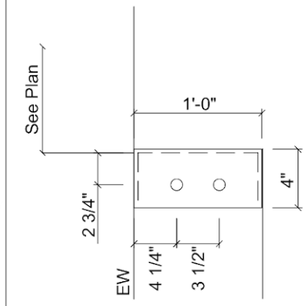
DETAIL C Base EL. 110'-0"

Dia= 3/4"



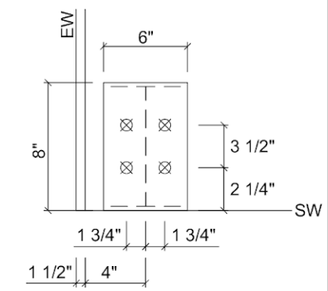
DETAIL D Base EL. 110'-0"

Dia= 5/8"



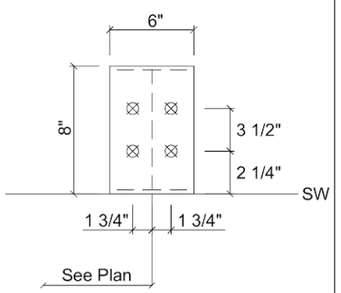
DETAIL E Base EL. 110'-0"

Dia= 3/4"



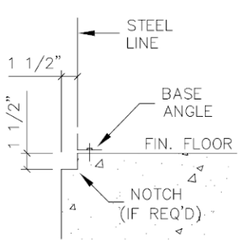
DETAIL F Base EL. 97'-10"

Dia= 3/4"

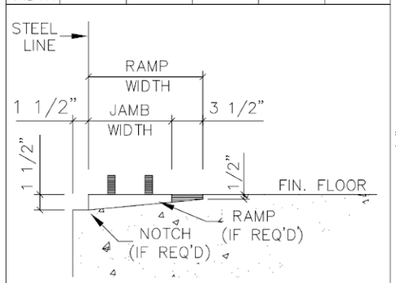


DETAIL G Base EL. 97'-10"

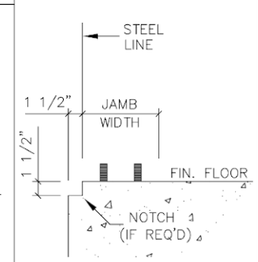
JAMB WIDTH	4"	6"	8"	10"	12"
RAMP WIDTH	7 1/2"	9 1/2"	11 1/2"	13 1/2"	15 1/2"



TYPICAL BASE SECTION



OVERHEAD DOOR BASE SECTION



FRAMED OPENING BASE SECTION

WEDGE ANCHOR BY OTHERS

Registered Professional Engineer (Structural) CERTIFICATE NO. 77126 COLEMAN D. LARSEN 01/30/24 Date Signed ARIZONA, U.S.A.

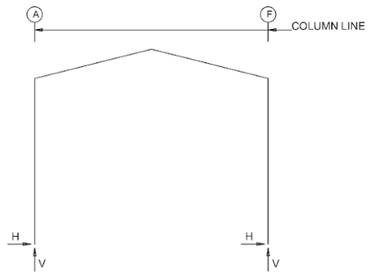
DRAWING REVISIONS	DRAWING USAGE
	For Construction 1/26/2024

IMAGE BUILDING SYSTEMS

	ANCHOR BOLT DETAILS		IM-0017-23
CUSTOMER	ADOT DE-ICING	PROJECT	ADOT DE-ICING
	SUPERIOR, AZ 85173		PRJ Address 1 SUPERIOR, AZ 85173

SHEET

FRAME LINES: 2 3 4



RIGID FRAME: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
2	A	3	2.0	3.9	6	-1.8	-2.0	4	0.750	6.000	12.00	0.375	120.0
		1	1.7	6.5	8	0.2	-3.9						
2	F	7	1.8	-2.1	2	-2.0	3.9	4	0.750	6.000	12.00	0.375	120.0
		1	-1.7	6.5	10	-0.2	-3.6						

RIGID FRAME: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
3	A	1	4.8	12.3	6	-3.2	-2.6	4	0.750	12.00	12.00	0.375	120.0
					4	-3.1	-5.0						
3	F	7	3.2	-2.6	1	-4.8	12.3	4	0.750	12.00	12.00	0.375	120.0
		1	-4.8	12.3	5	3.1	-5.0						

RIGID FRAME: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
4	A	1	1.8	5.2	6	-1.3	-1.0	4	0.750	6.000	12.00	0.375	120.0
					9	0.1	-2.4						
4	F	7	1.3	-1.0	1	-1.8	5.2	4	0.750	6.000	12.00	0.375	120.0
		1	-1.8	5.2	11	-0.1	-2.4						

NOTES FOR REACTIONS

1. All loading conditions are examined and only maximum/minimum H or V and the corresponding H or V are reported.
2. Positive reactions are as shown in the sketch. Foundation loads are in opposite directions.
3. Bracing reactions are in the plane of the brace with the H pointing away from the braced bay. The vertical reaction is downward.
4. Building reactions are based on the following building data:

Width (ft)	=	31.0
Length (ft)	=	48.0
Eave Height (ft)	=	22.0/ 22.0
Roof Slope (rise/12)	=	3.0/ 3.0
Dead Load (psf)	=	3.0
Collateral Load (psf)	=	3.0
Roof Live Load(psf)	=	20.0
Frame Live Load		
Min(psf)	=	12.0
Max(psf)	=	16.5
Snow Load (psf)	=	20.0
Wind Speed (mph)	=	103.0
Wind Code	=	IBC 18
Exposure	=	C
Closure	=	Enclosed
Importance Wind	=	1.00
Importance Seismic	=	1.00
Seismic Zone	=	C
Seismic Coeff (Fa*Ss)	=	0.45

5. Loading conditions are:

1 Dead+Collateral+Snow+Slide_Snow
2 Dead+Collateral+0.75Snow+0.45Wind_Left1+0.75Slide_Snow
3 Dead+Collateral+0.75Snow+0.45Wind_Right1+0.75Slide_Snow
4 0.6Dead+0.6Wind_Left1
5 0.6Dead+0.6Wind_Right1
6 0.6Dead+0.6Wind_Left2
7 0.6Dead+0.6Wind_Right2
8 0.6Dead+0.6Wind_Long1L
9 0.6Dead+0.6Wind_Long1R
10 0.6Dead+0.6Wind_Long2L
11 0.6Dead+0.6Wind_Long2R
12 0.6Dead+0.6Wind_Suction+0.6Wind_Long1L
13 0.6Dead+0.6Wind_Pressure+0.6Wind_Long1L
14 0.6Dead+0.6Wind_Right2+0.6Wind_Suction
15 0.6Dead+0.6Wind_Pressure+0.6Wind_Long2L
16 Dead+0.6Wind_Right2+0.6Wind_Suction

RIGID FRAME: BASIC COLUMN REACTIONS (k)

Frame Line	Column Line	-----Dead---- Horz	Vert	--Collateral- Horz	Vert	-----Live---- Horz	Vert	-----Snow---- Horz	Vert	--Wind_Left1- Horz	Vert	-Wind_Right1- Horz	Vert
2	A	0.3	1.0	0.2	0.7	1.0	4.0	1.3	4.7	-3.2	-6.1	1.3	-3.1
2	F	-0.3	1.0	-0.2	0.7	-1.0	4.0	-1.3	4.7	-1.3	-3.2	3.2	-6.2

Frame Line	Column Line	--Wind_Left2- Horz	Vert	-Wind_Right2- Horz	Vert	--Wind_Long1- Horz	Vert	--Wind_Long2- Horz	Vert	-Seismic_Left Horz	Vert	Seismic_Right Horz	Vert
2	A	-3.2	-4.4	1.3	-1.5	0.0	-7.5	-0.6	-6.3	-0.5	-0.4	0.5	0.4
2	F	-1.3	-1.6	3.2	-4.5	0.6	-6.8	0.0	-7.0	-0.5	0.3	0.5	-0.3

Frame Line	Column Line	-Seismic_Long Horz	Vert	F1UNB_SL_L- Horz	Vert	F1UNB_SL_R- Horz	Vert
2	A	0.0	-0.3	0.9	3.2	0.9	1.9
2	F	0.0	-0.3	-0.9	1.9	-0.9	3.2

Frame Line	Column Line	-----Dead---- Horz	Vert	--Collateral- Horz	Vert	-----Live---- Horz	Vert	-----Snow---- Horz	Vert	--Wind_Left1- Horz	Vert	-Wind_Right1- Horz	Vert
3	A	0.6	1.7	0.6	1.4	2.1	5.5	3.6	9.2	-5.8	-10.1	0.8	-6.3
3	F	-0.6	1.7	-0.6	1.4	-2.1	5.5	-3.6	9.2	-0.8	-6.3	5.8	-10.1

Frame Line	Column Line	--Wind_Left2- Horz	Vert	-Wind_Right2- Horz	Vert	--Wind_Long1- Horz	Vert	--Wind_Long2- Horz	Vert	-Seismic_Left Horz	Vert	Seismic_Right Horz	Vert
3	A	-5.9	-6.1	0.7	-2.4	0.3	-9.9	-0.6	-8.9	-0.4	-0.3	0.4	0.3
3	F	-0.7	-2.4	5.9	-6.1	0.6	-8.9	-0.3	-9.9	-0.4	0.3	0.4	-0.3

Frame Line	Column Line	-Seismic_Long Horz	Vert	F2UNB_SL_L- Horz	Vert	F2UNB_SL_R- Horz	Vert
3	A	0.0	-0.3	2.4	7.8	2.4	4.5
3	F	0.0	-0.3	-2.4	4.5	-2.4	7.8

Frame Line	Column Line	-----Dead---- Horz	Vert	--Collateral- Horz	Vert	-----Live---- Horz	Vert	-----Snow---- Horz	Vert	--Wind_Left1- Horz	Vert	-Wind_Right1- Horz	Vert
4	A	0.3	0.8	0.2	0.6	1.1	3.1	1.3	3.8	-2.3	-4.1	0.4	-2.6
4	F	-0.3	0.8	-0.2	0.6	-1.1	3.1	-1.3	3.8	-0.4	-2.6	2.3	-4.1

Frame Line	Column Line	--Wind_Left2- Horz	Vert	-Wind_Right2- Horz	Vert	--Wind_Long1- Horz	Vert	--Wind_Long2- Horz	Vert	-Seismic_Left Horz	Vert	Seismic_Right Horz	Vert
4	A	-2.4	-2.5	0.4	-1.0	0.0	-4.8	-0.6	-4.1	-0.2	-0.2	0.2	0.2
4	F	-0.4	-1.0	2.4	-2.5	0.6	-4.1	0.0	-4.8	-0.2	0.2	0.2	-0.2

Frame Line	Column Line	F3UNB_SL_L- Horz	Vert	F3UNB_SL_R- Horz	Vert
4	A	0.9	3.2	0.9	1.9
4	F	-0.9	1.9	-0.9	3.2

ENDWALL COLUMN: BASIC COLUMN REACTIONS (k)

Frm Line	Col Line	Dead Vert	Collat Vert	Live Vert	---Snow--- Horz	Vert	Wind Left1 Vert	Wind Right1 Vert	Wind Left2 Vert	Wind Right2 Vert	Wind Press Horz	Wind Suct Horz	Wind Long1 Vert
2	D	0.6	0.3	2.3	-0.1	4.7	-2.2	-2.2	-2.2	-2.2	-4.3	5.3	-2.8
4	C	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-2.1	2.4	0.0

Frm Line	Col Line	Wind Long2 Vert	Seis Long Vert
2	D	-1.1	0.0
4	C	0.0	0.0

ENDWALL COLUMN: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
2	D	12	3.2	-1.3	13	-2.6	-1.3	4	0.625	6.000	12.00	0.375	120.0
		1	-0.1	5.7	12	3.2	-1.3						
4	C	14	1.4	0.1	15	-1.3	0.1	4	0.625	6.000	8.000	0.375	120.0
		16	1.4	0.1									

ANCHOR BOLT SUMMARY

	Qty	Locate	Dia (in)	Type
○	4	Jamb	5/8"	
○	8	Endwall	5/8"	A307
⊠	24	Frame	3/4"	A307

BUILDING BRACING REACTIONS

Wall Loc	Line	Col Line	± Reactions(k) Wind Horz	Vert	Seismic Horz	Vert	Panel_Shear (lb/ft) Wind	Seis	Note
L_EW	2								(h)
F_SW	F	2,3	3.0	1.3	0.8	0.3			
R_EW	4								(h)
B_SW	A	3,2	3.0	1.3	0.8	0.3			

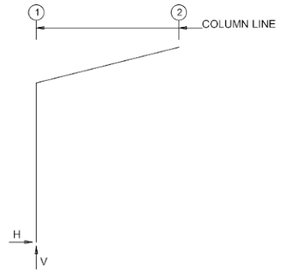
(h)Rigid frame at endwall

Reactions for seismic represent shear force, Eh

Registered Professional Engineer (Structural) CERTIFICATE NO. 77126 COLEMAN D. LARSEN 01/30/24 Date Signed ARIZONA, U.S.A.

DRAWING REVISIONS	DRAWING USAGE				
	For Construction 1/26/2024	IMAGE BUILDING SYSTEMS	ANCHOR BOLT REACTIONS	MAIN	Page 5 of 24
			CUSTOMER: ADOT DE-ICING	PROJECT: ADOT DE-ICING	SHEET
			SUPERIOR, AZ 85173	SUPERIOR, AZ 85173	

FRAME LINES: E D B



RIGID FRAME: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
E	1	5	0.5	-0.6	3	-0.5	0.3	4	0.750	6.000	8.000	0.375	0.0
		1	0.0	1.7	4	0.5	-0.8						

RIGID FRAME: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
D	1	5	1.2	-1.3	3	-1.3	0.5	4	0.750	6.000	8.000	0.375	0.0
		1	0.1	3.9	4	1.2	-1.6						

RIGID FRAME: MAXIMUM REACTIONS, ANCHOR BOLTS, & BASE PLATES

Frm Line	Col Line	Column_Reactions(k) Load Id	Hmax H	V Vmax	Load Id	Hmin H	V Vmin	Bolt(in) Qty	Dia	Base_Plate(in) Width	Length	Thick	Grout (in)
B	1	5	0.5	-0.5	3	-0.5	0.3	4	0.750	6.000	8.000	0.375	0.0
		1	0.0	1.7	4	0.5	-0.8						

BUILDING BRACING REACTIONS

Wall Loc	Wall Line	Col Line	± Reactions(k) Wind Horz	Wind Vert	Seismic Horz	Seismic Vert	Panel_Shear (lb/ft) Wind	Seis	Note
L_EW	E								(h)
F_SW	2								(e)
R_EW	B								(h)
B_SW	1	Torsional Bracing Used							

(e)Bracing loads must be applied to supporting building
(h)Rigid frame at endwall

Reactions for seismic represent shear force, Eh

RIGID FRAME: BASIC COLUMN REACTIONS (k)

Frame Line	Column Line	-----Dead---- Horz	Vert	--Collateral- Horz	Vert	-----Live---- Horz	Vert	-----Snow---- Horz	Vert	--Snow_Drift- Horz	Vert	--Wind_Left1- Horz	Vert
E	1	0.0	0.3	0.0	0.1	0.0	1.0	0.0	1.0	0.0	0.3	0.0	-1.1
D	1	0.0	0.5	0.0	0.4	0.0	2.3	0.0	2.3	0.0	0.7	0.1	-2.7
B	1	0.0	0.3	0.0	0.1	0.0	1.0	0.0	1.0	0.0	0.3	0.0	-1.1

Frame Line	Column Line	-Wind_Right1- Horz	Vert	--Wind_Left2- Horz	Vert	-Wind_Right2- Horz	Vert	--Wind_Long1- Horz	Vert	--Wind_Long2- Horz	Vert	-Seismic_Left Horz	Vert
E	1	0.8	-1.1	-0.9	0.1	-0.2	0.1	0.9	-1.7	0.9	-1.3	0.0	0.0
D	1	1.9	-2.7	-2.2	0.3	-0.4	0.4	2.0	-3.2	2.1	-2.7	0.0	0.0
B	1	0.8	-1.1	-0.9	0.1	-0.2	0.1	0.9	-1.7	0.9	-1.2	0.0	0.0

Frame Line	Column Line	Seismic_Right Horz	Vert	-Seismic_Long Horz	Vert
E	1	0.0	0.0	0.0	0.0
D	1	0.0	0.0	0.0	0.0
B	1	0.0	0.0	0.0	0.0

ANCHOR BOLT SUMMARY

	Qty	Locate	Dia (in)	Type
☒	12	Frame	3/4"	A307

NOTES FOR REACTIONS

1. All loading conditions are examined and only maximum/minimum H or V and the corresponding H or V are reported.
2. Positive reactions are as shown in the sketch. Foundation loads are in opposite directions.
3. Bracing reactions are in the plane of the brace with the H pointing away from the braced bay. The vertical reaction is downward.
4. Building reactions are based on the following building data:

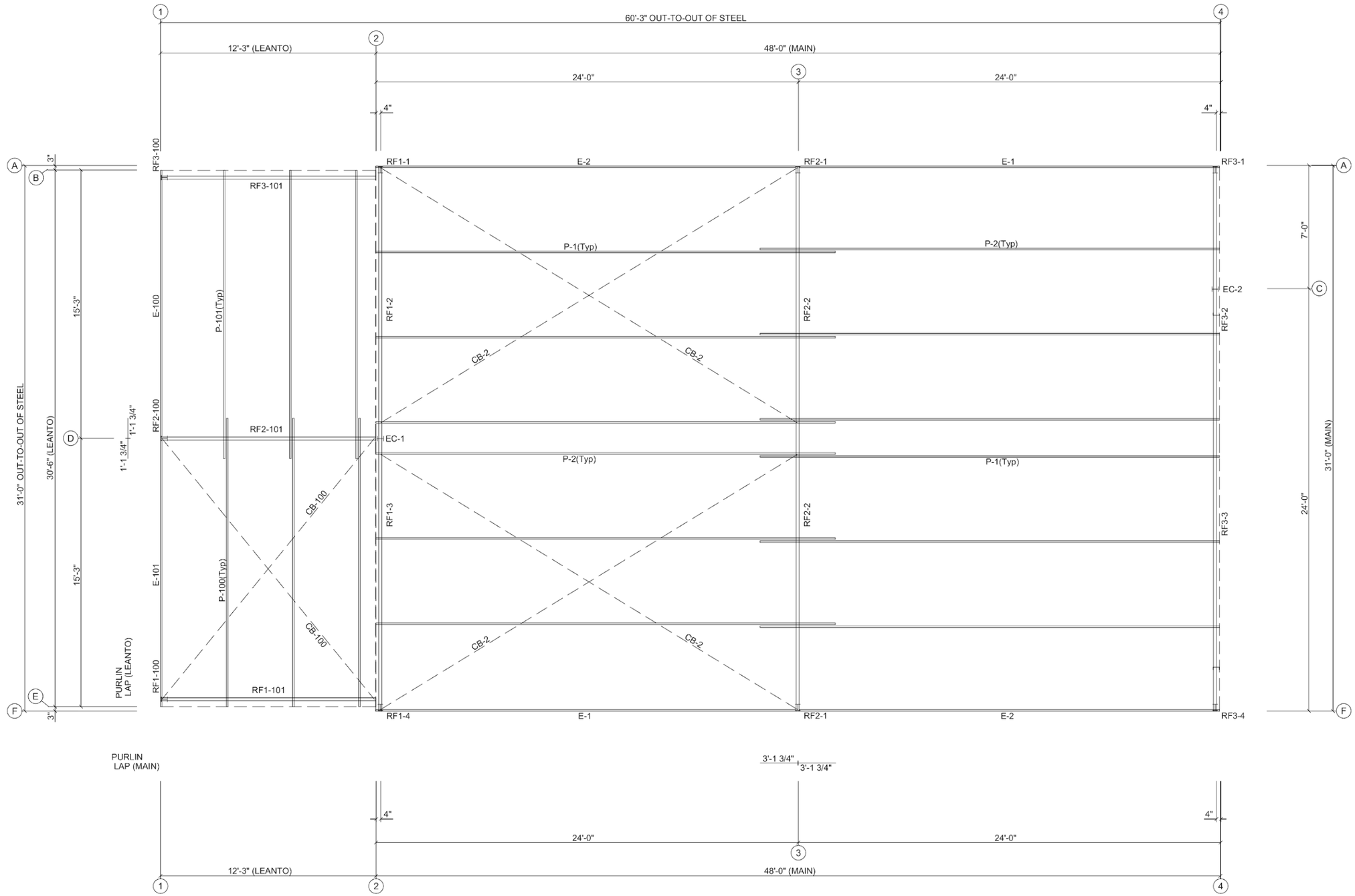
Width (ft)	=	12.3
Length (ft)	=	30.5
Eave Height (ft)	=	13.7/ 16.8
Roof Slope (rise/12)	=	3.0
Dead Load (psf)	=	3.0
Collateral Load (psf)	=	3.0
Live Load (psf)	=	20.0
Snow Load (psf)	=	20.0
Wind Speed (mph)	=	103.0
Wind Code	=	IBC 18
Exposure	=	C
Closure	=	Partially Enclosed
Importance Wind	=	1.00
Importance Seismic	=	1.00
Seismic Zone	=	C
Seismic Coeff (Fa*Ss)	=	0.45

5. Loading conditions are:
   1 Dead+Collateral+Snow+Snow_Drift
   2 0.6Dead+0.6Wind_Left1
   3 0.6Dead+0.6Wind_Left2
   4 0.6Dead+0.6Wind_Long1R
   5 0.6Dead+0.6Wind_Long2R

Registered Professional Engineer (Structural) CERTIFICATE NO. 77126 COLEMAN D. LARSEN 01/30/24 Date Signed ARIZONA U.S.A.

DRAWING REVISIONS	DRAWING USAGE		ANCHOR BOLT REACTIONS	LEANTO		
	For Construction 1/26/2024	IMAGE BUILDING SYSTEMS	CUSTOMER: ADOT DE-ICING	PROJECT: ADOT DE-ICING	SHEET	Page 6 of 24
			SUPERIOR, AZ 85173	SUPERIOR, AZ 85173		

MEMBER TABLE: FRAME LINE			
QUAN	MARK	PART	LENGTH
6	P-1	8X25Z12	27'-1 1/2"
6	P-2	8X25Z12	27'-1 1/2"
2	E-1	E085343L	23'-11 1/2"
2	E-2	E085343L	23'-11 1/2"
4	CB-2	HW373	28'-3 1/2"
3	P-100	8X25Z16	16'-4 1/2"
3	P-101	8X25Z16	16'-4 1/2"
1	E-100	E085343L	15'-2 1/2"
1	E-101	E085343L	15'-2 1/2"
2	CB-100	HW373	19'-0"



ROOF FRAMING PLAN

Registered Professional Engineer (Structural) CERTIFICATE NO. 77126 COLEMAN D. LARSEN 01/26/24 Date Signed ARIZONA, U.S.A.

DRAWING REVISIONS	DRAWING USAGE				
	For Construction 1/26/2024	IMAGE BUILDING SYSTEMS	ROOF FRAMING	IM-0017-23	Page 7 of 24
			CUSTOMER: ADOT DE-ICING	PROJECT: ADOT DE-ICING	SHEET
			SUPERIOR, AZ 85173	PRJ Address 1 SUPERIOR, AZ 85173	